Session-X: Alternate irrigation practices: December 2nd week

Program schedule

S.No	Time	Content	Method	Materials	Expected outcomes
1	15 min.	Recap.	Large group discussion	Charts and markers.	Participants recollect the contents discussed in ninth session.
2	45 min.	LTE observation and analysis.	Small group discussion	FWS group kit, charts and markers.	Participants observe the changes in experimental and control plots and analyze the data.
3	60 min.	Estimation of water requirement: Conventional Vs. alternate irrigation practices.	Small group discussion	Charts, markers and calculators.	Participants discuss: • Conventional methods of irrigation and estimate water requirement. • Alternate methods of irrigation and estimate water requirement. • Perceived change.
4	15 min.	Group dynamics- Linkages with line departments.	Role play	Script for role play.	Participants identify the need to have good relations with government and other agencies for sustainability of HUNs and GMCs.
5	20 min.	Short Study- Alternate furrow irrigation method.	Demonstration	Plastic bottles, and chilli plants.	Participants observe the difference in water requirement between traditional method and alternate furrow irrigation method in Chilli crop.
6	5 min.	Homework: Interaction with the farmers who have been practicing alternate irrigation techniques.	Large group	Handout, Charts, Markers.	Participants will interact with other farmers in the village to know more about alternate irrigation techniques.
7	10 min.	Evaluation.	Large group discussion	Charts and markers.	Participants share their opinions on the conduct of FWS sessions and methods used in the session.

Learning objectives

By the end of the session, participants will be able to:

- Recap the previous session,
- Observe and analyze the long term experiment,

- Estimate the quantity of water required under conventional vs. alternate irrigation practices.
- Discuss the importance of having linkages with line departments.
- Demonstrate the impact of organic matter on water holding capacity in different soils.

Process

1. Recap Time: 15 min.

Objective : Participants recap the contents discussed in the previous session.

Content : Recap session nine contents

- Observation of LTE and data collection,
- Sharing crop adoption survey Results,
- Estimation of actual water requirement for the HU.
- Importance of collective decision making.

Method : Large group discussion

Materials : Charts and markers.

Process

- Welcome participants and start the session with FWS pledge.
- In the large group, encourage each participant to share a content that was discussed in previous session and the learning from that one.
- As you facilitate the recap, ask one of the participants to list the contents on the chart paper.
- Discuss the contents and the learning from the previous session.

2. LTE observation and analysis Time: 45min.

Objective : Participants observe and analyze the Long Term Experiment (LTE).

Content :

- Observation of LTE, and
- Data analysis.

Method : Small group discussion

Materials : FWS group kit, charts and markers.

Process

- Request participants to visit the filed and note their observations.
- Encourage the participants note their observations during the filed visit.

Observation sheet:

Village: Name of experiment:

Farmer: Crop:

Area:

Date	Time required per irrigation	Time taken for filling 100 lit drum (Sec, min and 1 hour)	Volume of water	Draft	Yield	Observation

- Ask the groups to come back to the training spot and analyze the observations made during the field visit. Request each group to use HESA (based on the experiment) frame work to present their group work.
- Request collaborator to implement the decisions taken by the large group in the experimental plot.

3. Estimation of water requirement: Conventional Vs. Alternate irrigation practices

Time: 60 min.

Objective : Participants discuss:

- Conventional methods of irrigation and estimate water requirement.
- Alternate methods of irrigation and estimate water requirement.
- Discuss perceived change.

Content :

- Conventional methods of irrigation.
- Alternate method of irrigation.
- Estimation of water requirement.

Method : Large group discussion, and small group discussion.

Materials : Charts, markers and calculators.

Process

Conventional methods of irrigation and water requirement:

- Ask participants the following questions in large group.
  - What are the crops grown in the hydrological unit?
  - What are the conventional irrigation practices for each crop?
- Write the responses on a chart.
- Ask a participant to explain the process of estimating water requirement for a particular crop. Remind participants about the estimation of water requirement for a crop.
- Request participants to work in groups and assign one or two crops based on the number of crops in the HU, and corresponding conventional irrigation practice to each group.
- Ask each group to estimate the quantity of water required for the crops. Provide the value of average discharge to each group.
- Encourage the groups to report their group work to the larger group.
- Summarize the discussions and list the contents in the following table.

Crop	Method of irrigation	Water required (Ltrs.)

Alternate methods of irrigation and water requirement:

- Ask the participants in large group to list alternate irrigation practices for the identified crops.
- List the responses on the chart.

- Ask participants to continue working in the same groups. Assign one or two crops and relevant alternate irrigation practice to each group and ask them to estimate the quantity of water required to raise the crop.
- Provide the value of average discharge of the bore well each group.
- Ask the groups to present their group work to large group.
- Summarize the discussions using the following table.

Crop	Alternate irrigation practice	Water required (Ltrs.)

- Compare the two tables and discuss the differences in quantity of water required for the same crop.

4. Group dynamics: Linkages with line departments Time: 15 min

Objective : Participants understand the importance of linkages with line departments.

Content : Linkages with line departments.

Method : Role play

Materials : Script for role play.

Process

- Ask three participants to volunteer for a short role play.
- Inform that as a part of role play, these three participants will be meeting with the agriculture officer (facilitator will enact this role) to represent their HUN/GMC and discuss the subsidy for drip and sprinklers.
- Give the three participants few minutes to discuss how they wish to approach and discuss the issue with the officer.
- Meanwhile, share with the rest of the participants the following checklist. Ask them to use the checklist to evaluate the interaction in the role-play.
    - Prior consultation amongst members about the subsidy.
    - An application to the officer on the GMC/HUN letter pad.
    - Members introduce themselves and their roles in the GMC/HUN to the officer.
    - Members are confident in their interactions with the agriculture officer politely.
    - One person speaks at a time.
    - Ask relevant questions to gather necessary information.
    - Thank the officer as they leave the meeting.

5. Short study: Alternate furrow irrigation method Time: 20 min**.**

Objective : Participants will observe the difference in water requirement between traditional method and alternate furrow irrigation method in Chilli crop.

Rationale : Demonstrating the difference between traditional method and alternate furrow method of irrigation in chilli crop using a model will help the farmers understand the quantity of water saved in alternate furrow method of irrigation**.**

Materials : Two plastic bottles of two litre capacity to water two Chilli plants.

Usage

- Take two empty plastic bottles and mark the readings from bottom to top with 10 - 20 - 30 - 40 - 50 numbers.
- Name the first bottle as traditional method of irrigation and the second bottle as alternate furrow method of irrigation.
- Place a chilli plant in first bottle and pour water up to 30 mark indicating that traditional method of irrigation requires 30 lakh liters of water.
- Place another Chilli plant in second bottle and pour water up to 20 mark indicating that alternate furrow method of irrigation requires 20 lakh liters of water
- Discuss the difference in water requirements for Chilli crop between the two methods of irrigation.

6. Homework: Interaction with farmers practicing alternate irrigation techniques

Time: 5 min.

- Request farm participants to discuss about useful ness and efficiency of alternate irrigation techniques who have been practicing in the village or neighboring villages

7. Evaluation Time: 10 min.

Ask the following questions to evaluate the day's learning. Pose the questions to the large group:

- What are alternate methods of irrigation?
- How can we increase water holding capacity of soils?
- Which soil has maximum water holding capacity and why?

Encourage participants to share their learning and thank the participants for their participation in the session.